

Key Performance Indicators for BucStop Usability and Team Velocity

Author(s): Zoe & Grace

Abstract

This report outlines the key performance indicators (selected for evaluating the BucStop application and development process. The selected metrics aim to provide insight into system performance, user engagement, and team productivity.

1. Introduction

Metrics provide essential feedback for continuous improvement in software projects. For BucStop, a lightweight project developed in a team environment, we have identified three KPIs that balance practicality with meaningful insights.

2. Selected KPIs and Rationale

2.1 Time to Process a Submission

This KPI measures the system's responsiveness by capturing the duration from when a user clicks the "Submit" button to when the corresponding data is successfully written to persistent storage. This metric will be recorded using backend logging of timestamped console output.

2.2 Actual Velocity (Team Hours Worked)

Velocity reflects the total number of hours the team spends on sprint-related development tasks, excluding planning and meeting time. It helps gauge the

realistic effort behind sprint completions and allows for future sprint planning improvements.

2.3 Game Completion Rate

To better understand user engagement and the effectiveness of game design, we track the ratio of game sessions started to those completed (defined as reaching a final score screen). A high drop-off rate may indicate confusing instructions or an unengaging user experience.

3. Conclusion

These metrics provide visibility into system performance and team effort, which are vital for identifying bottlenecks and improving usability. Given that BucStop is a low-functioning application intended to host quick mini-games for use between classes — and many of the games currently do not function these KPIs offer a practical foundation for diagnosing issues, prioritizing improvements, and tracking basic development progress. The submission time helps ensure backend logic is functioning as expected, the velocity metric keeps the team accountable for steady progress, and the game completion rate sheds light on how playable and engaging the games are, even in early stages.